

It is not required that your answers be submitted *on* this worksheet, but you may choose to do so; if you do, this pdf has been designed so that you can select “Assign Pages Sequentially” in Gradescope. If you choose to submit in another format, then: when submitting your questions, be sure that

**each answer is submitted to the correct question**

(you can change the order of pages within GradeScope) and that the page is

**oriented correctly**

(you can rotate the pages within GradeScope). Otherwise,

**your question will not be graded.**

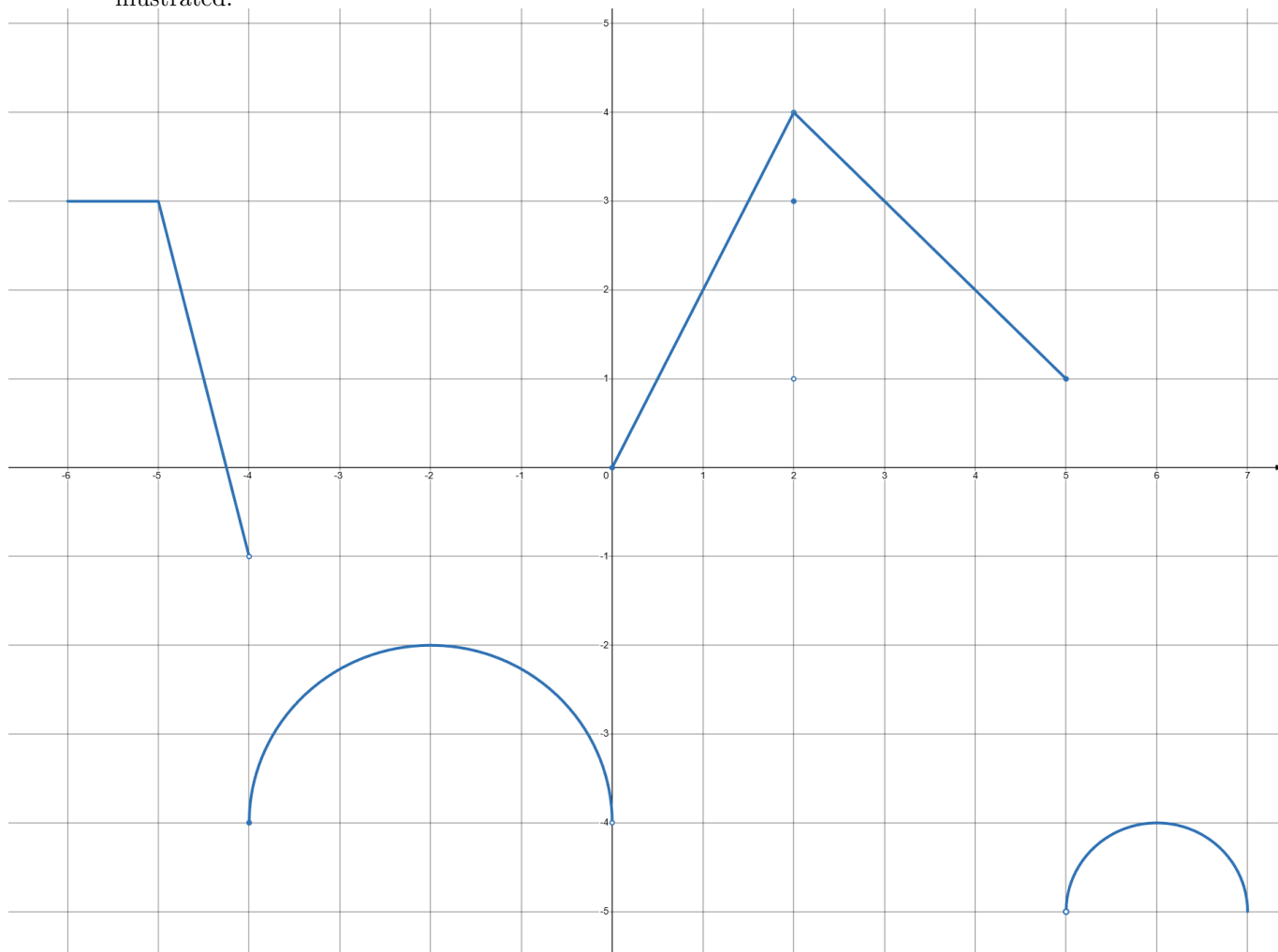
Your work may be submitted on your own paper, by writing on this worksheet printed out, or by writing via tablet, among possibly other options. If you are having problems submitting your work, please attend office hours. (There is no late penalty for submitting the assignment on the due date, but there will be no late submissions accepted for any reason.) Click on [Gradescope's help page](#) and read their instructions.

1. Find the derivatives of the following functions.

(a)  $f(x) = \int_{-3\pi}^{\sin(6x)} \ln(1+t^4) dt$

(b)  $h(t) = \int_t^{\sqrt{t}} (\sin(\theta^8))^{2026} d\theta$

2. The graph of a function  $f$ , consisting of only line segments and portions of circular arcs is illustrated.



Define  $g(x) = \int_2^x f(t) dt$ .

- Using just geometry, compute  $g(3)$ ,  $g(0)$ , and  $g(-4)$ . Include work.
- Find the domain (expressed in interval notation) on which  $g$  is increasing. Explain your reasoning.
- Find the  $x$ -coordinate of the absolute maximum of  $g$  on the interval  $[-6, 7]$ . Be sure to justify.

3. Find the specified antiderivatives (or integrals). You must include work.

(a)  $\int x^3 \sin(x^4) \cos^7(x^4) dx$

(b)  $\int_0^1 x^{3/2} e^{\sqrt{x}} dx$

(c)  $\int \sin^9 \theta \cos^4 \theta \, d\theta$

(d)  $\int \frac{2x}{(x^2 + 1)^{3/2}} \, dx$

No credit: It is possible to handle  $\int \frac{2x}{(x^2 + 1)^{3/2}} \, dx$  using a trigonometric substitution.

No credit: It is also possible to handle  $\int \frac{2x}{(x^2 + 1)^{3/2}} \, dx$  using the hyperbolic substitution  $x = \sinh(t)$ .